

Applications of Thermal Processes

Conduction

1. Hair dryers

- Hair dryers make use of fast flowing hot air to dry out hair quickly
- A motor pumps cool air through a heating element
- energy is transferred from heating element to fast flowing air via conduction

2. Instant water heaters

- Instant water heaters employ a similar mechanism as hair dryers to heat water quickly.
- Cold water is passed through heating elements that transfer energy to water via conduction.

3. Thermal wear

- Thermal wear contains porous materials like down feathers and microfibers that trap tiny pockets of air.
- As air is trapped in constrained space, there is no convection current.
- trapped air in thermal wear acts as an insulating layer and reduces body heat to surroundings via conduction.

4. Double/Triple glazed windows

- air is a poor thermal conductor. thin layer of air trapped between 2 glass panes reduces energy transfer via conduction through window
- Energy transfer through window is further reduced with 3 glass panes and 2 layers of trapped air.
- Double or triple glazed windows help to keep interior of the home cool during summer and keep interior warm during winter.

Convection

1. Electric kettles

- Heating elements are usually placed at bottom of electric kettles.
- when kettle is switched on, water around heating element heats up and expands.
- Heated water rises while cooler water at top part of the kettle sinks to replace heated water.
- convection current is set up - which allows water in kettle to be heated up more quickly and evenly.

2. Hot air balloons.

- Burner at bottom of a hot air balloon heats up air above it.
- heated air expands and rises to the top of the balloon and displaces the cooler air.
- cooler air sinks to the bottom of the balloon, where it gets heated up. this movement of air sets up convection current.
- hot air has lower density than cool air. when sufficient air in the balloon is heated up, balloon rises.

3. Sea Breeze

- During day, land heats up faster compared to sea water
- air above land heats up and rises.
- cooler air above sea rushes inland to replace space left by warm air
- sets up convection current, creating sea breeze.

4. Land breeze

- during night, land cools down faster compared to seawater
- air above the sea heats up and rises.
- cooler air above the land rushes out towards sea to replace space left by warm air.
- this sets up a convection current, creating land breeze.

Radiation

1. Greenhouses

- suns radiation passes thru the glass and heats up the ground in greenhouse
- ground in the greenhouse becomes warm and re radiates infrared radiation with longer wavelengths.

- infrared radiation with longer wavelengths are unable to pass through the glass roof. it gets reflected and trapped in the greenhouse. this keeps greenhouse warm to support plant growth.

2. Space Blanket

- Space blanket has shiny and smooth surface
- useful in emergency situations such as to help person who has fallen into very cold water, warm as it is a poor emitter of infrared radiation.
- can be used to wrap up houses to protect the houses against large scale bushfires as it is a poor absorber of infrared radiation. prevents the radiation from reaching and damaging the houses.

3. Global warming

- earths atmosphere → is like a greenhouse.
 - solar radiation that enters the atmosphere warms earths surface.
 - surface then re radiates infrared radiation at longer wavelengths that are more easily absorbed by atmospheric greenhouse gasses.
 - radiation absorbed will escape into outer space while remaining will have to go through more cycles of re absorption and re radiation before it escapes into space.
 - amount of greenhouse gasses in the atmosphere has increased drastically due to human activities such as burning of fossil fuels.
 - thus more energy is trapped by greenhouse gasses and temperature on earth increases, leading to global warming.
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